Alden Grover

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Summary

Seeking spring/summer 2027 internships in digital circuit and RTL design, verification, and research. BS/MS combined student with Ph.D. intent, open to industry roles between degrees.

Education

Carnegie Mellon University, Pittsburgh, PA
BS in Electrical and Computer Engineering, expected graduation December 2026
Expected MS in Electrical and Computer Engineering (combined MS/BS program), graduating 2028, with Ph.D. intent
Honors: College of Engineering Dean's List (Spring 2024)
Notable Classwork: 18-447: Introduction to Computer Architecture, 18-320: Microelectronic Circuits, 18-340: Hardware Arithmetic for Machine Learning, 18-349: Introduction to Embedded Systems

Technical Skills

- **RTL Design:** SystemVerilog, Verilog, ASIC RTL design and tape out, FPGA design, TinyTapeout, network-on-chip (NoC) design, USB 2.0 hardware, finite state machines (FSMs) for datapath control, VGA waveform generation
- **RTL Verification and Simulation:** SystemVerilog testbenches, Cocotb, RTL waveform debug, formal reachability proofs for FSMs, static timing analysis (STA), constrained random testing
- **EDA and Flow:** Cadence Virtuoso, Cadence synthesis and simulation, Yosys, SPICE, RTL-to-GDSII flow (netlist, place and route, GDSII), TSMC 28nm planar PDK, Sky130 PDK, generic 45nm planar PDK, KiCad
- **Computer Architecture:** RISC-V (RV32I/M) ISA and microcode, pipelining, branch prediction, cache management and bus snooping
- **Clocking and Custom Circuits:** PLLs, ring oscillators, digitally controlled delay lines (DCDL), standard-cell library design, PPA (power, performance, area) optimization
- **Embedded Systems:** Embedded C++, RTOS development, bootloader design, UART/USART and QSPI, custom serial protocols, hard real-time systems, memory-mapped peripheral firmware development, sensor fusion (Kalman filter)
- **Hardware Acceleration:** Systolic arrays, neural-network inference, KNN acceleration, decision-tree acceleration, floating-point arithmetic
- **Languages:** C, C++, SystemVerilog, Python, Julia
- **Tools:** Git and GitHub, PCB design, CAD, CAM

Research Experience

Design space exploration for Pareto-optimal design of a Digitally Controlled Delay Line utilizing current starved and path choice methods
Ken Mai and Jim Bain Research Group, Carnegie Mellon University — Aug 2026 - Present (in preparation)

- Leading investigation into a Pareto-optimal method to combine path choice and current starved Digitally Controlled Delay Line techniques.
- Investigating impact of existing implementations of path choice and current starved DCDL lines on range/resolution per unit area for PVT corners.
- Developing techniques for determining the optimal combination of path choice and current starved implementations.
- Verifying proposed techniques with brute force simulation on a small-scale model and directed simulation on a large-scale implementation.

IC I/O Harness on Commercial Processes and Tools
Ken Mai and Jim Bain Research Group, Carnegie Mellon University — Nov 2025 - Present

- Designed a portion of a taped-out ASIC with a focus on PLL, ring oscillators, and FPGA fabric in the TSMC 28nm PDK.
- Designed a ring oscillator and digital phase-locked loop that generates a 320 MHz on-chip clock from a 20 MHz reference, a 16-fold increase.
- Redesigning the PLL for a new tape-out, targeting a 400 MHz on-chip clock frequency.
- Verifying PLL behavior with SPICE simulation.
- Designed custom FPGA fabric to interface with the I/O Harness, helping students familiarize themselves with it without the complicated pitfalls of ASIC design.
- Designed a PCB carrier board to interface the finalized ASIC with an external FPGA for bring-up and post-silicon validation.
- **Technologies:** RTL design in SystemVerilog, synthesis and simulation with Cadence.

Nanopositioning System for Lithography and Van der Waals Heterostructure Construction
HackerFab, Carnegie Mellon University — Sep 2025 - Dec 2025

- Led a team that improved high-accuracy nanopositioning system resolution from 10μm to 400nm.
- **Technologies:** Embedded C++, CAD and CAM, piezoelectric nanopositioning.

Leadership

Class Teacher, 98-341: Build Your Own Breadboard Computer
Carnegie Mellon University — Aug 2026 - Present

- Teaching 98-341, covering the design and construction of a breadboard computer.

Teaching Assistant, 98-341: Build Your Own Breadboard Computer
Carnegie Mellon University — Jan 2026 - May 2026

- Supported students with hardware debugging and design questions in the same course.

Electronics Master, Robotics Club — Aug 2025 - Present

- Lead the electronics subgroup of the club.

Project Experience

Standard Cell Library Design — Apr 2026 - Present

- Designed a standard cell library in a 45nm planar PDK.
- Extracted parasitic behavior for use in timing characterization.
- Verified functionality and characterized performance with SPICE simulation.
- Cell library consisted of NAND, NOR, and NOT gates as well as DFF and latch circuits.

Physics Engine ASIC — Dec 2025 - Present

- Tape out: Sky130 open-source PDK (SkyWater 130nm), via the TinyTapeout shuttle.
- Implemented a falling sand simulation with output over VGA.
- QSPI memory controller reduced required on-chip memory to 1/8 of original size (from 96 bytes to 12 bytes).
- Complete ASIC design flow: RTL design, RTL simulation/debug, netlist generation, place and route, GDSII generation.
- Reduced clock cycles per physics simulation tick by a factor of three by adapting a physics simulation technique from graphics programming.
- Utilized RTL waveform debugging to fix bugs in the RTL design.
- Interfaced with external RAM hardware via QSPI, requiring intra-clock edge timing control.
- Leveraged fixed-delay digital buffers to ensure all protocols are obeyed.
- Constructed a test bench using Cocotb and Python, and verified functionality with assertions and constrained random testing.

Lightweight Autonomous Vehicle Firmware, Robobuggy Jan 2023 - Present

- Designed, constructed, and maintained a high-performance embedded system in C++ over the course of four years. Lead designer for two of those years.
- Leveraged an existing embedded system architecture and carefully chosen consumer hardware to maximize performance per dollar.
- Embedded software engineering with hard real-time requirements and tight control loops. Average control loop time under 10 microseconds.
- Designed PCBs to interface microcontrollers with peripheral devices including GPS, servo actuators, accelerometers, encoders, and 433MHz radios.
- Collaborated with control-system experts to design an Unscented Kalman Filter to integrate disparate sensor streams into a cohesive whole.

Static Timing Analysis Tool Jan 2026 - May 2026

- Built a static timing analysis tool for the Yosys Open Synthesis Suite.
- Able to determine timing behavior of non-combinational loop netlists.
- Timing calculations using PVT corners ensure accuracy of results in all expected environmental conditions.

7-stage Patterson and Hennessy style RV32I Core Jan 2026 - May 2026

- Implemented an RV32I core in SystemVerilog with 7 pipeline stages.
- Maximal forwarding between pipeline stages minimized data-dependent stalling. The final implementation was more than 5 times faster than the naive approach using real-world RV32I programs.
- Designed a two-bit branch predictor with a branch target buffer that doubled accurate branch predictions from 40% to 80%.
- Implemented a cache-management solution that successfully arbitrated cache conflicts inherent in non-atomic bus snooping systems.

FP32 FPU and Neural Network Systolic Array Sep 2025 - Dec 2025

- Designed IEEE 754 compatible hardware for computing 32-bit floating point operations.
- Supports all arithmetic operations (add, subtract, multiply, divide).
- Performed mathematical modeling to optimize the addition system for power, performance, and area.
- Custom multiplication and division architecture enabled single-cycle operation and low-impact change to a highly pipelined design.
- Implemented the FP unit in a systolic array used to accelerate inference of a deep neural network.

K Nearest Neighbor Inference Accelerator Sep 2025 - Dec 2025

- Designed a series of hardware threads that implemented the K Nearest Neighbor machine learning algorithm.
- Heavily pipelined design and Manhattan distance calculation maximized clock frequency.
- Exceeded the 300 MHz timing requirement (PDK specified by 18-340).

Decision Tree Training Accelerator Sep 2025 - Dec 2025

- Designed a series of hardware threads that accelerated decision tree machine learning algorithm training.
- Calculated Gini score using custom multiplication hardware.
- Leveraged arithmetic properties of numerical representation to bypass expensive computations.
- Custom multiplication hardware reduced cycles per operation from 32 to 1.

Network-on-a-Chip Jan 2025 - May 2025

- Designed and simulated four port packet routing hardware supporting interconnected routers and fair packet flow between ports.
- Parallel FIFO buffers and internal counters ensure high throughput despite large data flow.
- Address logic enabled packet transmission through multiple routers to reach the destination.
- Internal data representation minimized packet latency through the router.

USB 2.0 Hardware Jan 2025 - May 2025

- Implemented read and write operations.
- 8-way pipelined and parallelized design enables rapid data transmission and reception, 8 times faster than a serial design.
- Custom orchestration hardware enables device-to-device transfers without a host OS.

Matrix Multiplication, Addition, and Inner Product Hardware Sep 2024 - Dec 2024

- Interfaced with DSP slices and ROMs on an Intel Altera DE10 FPGA.
- Coordination between all available DSP hard cores and ROM modules within 98% of the dataflow limit on the number of compute cycles.

CPU Arithmetic Coprocessor and Microcode Modification Sep 2024 - Dec 2024

- Modified a multicycle CPU to support double-word arithmetic operations via microarchitecture changes to interface with an arithmetic coprocessor.
- Coprocessor design minimized instruction decode logic and microcode architecture, and eliminated software-based data dependence hazards using hardware solutions.

Real-Time Operating System and Bootloader for ESP32 Jan 2024 - May 2024

- Wrote bootloader and fixed priority preemptive scheduling RTOS for ESP32.
- Developed firmware for memory mapped and GPIO-based peripherals including USART/UART, H-Bridge BLDC motor controllers, rotational encoders, and LCD displays.